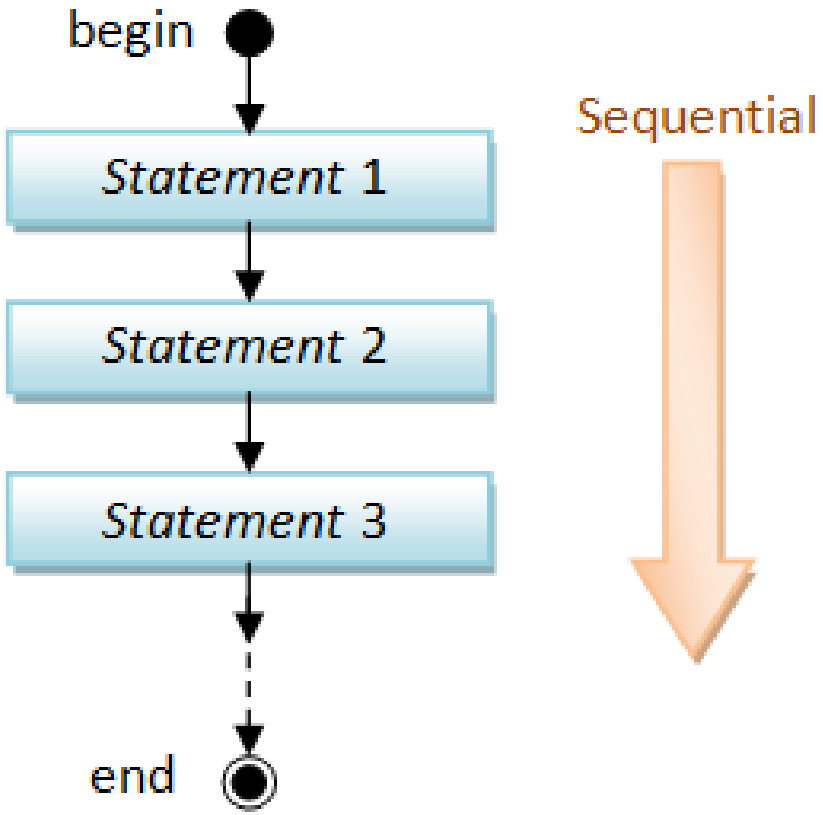
Control Structure

Control Structure

- Sequence
- Selection
- Repetition
- Function call

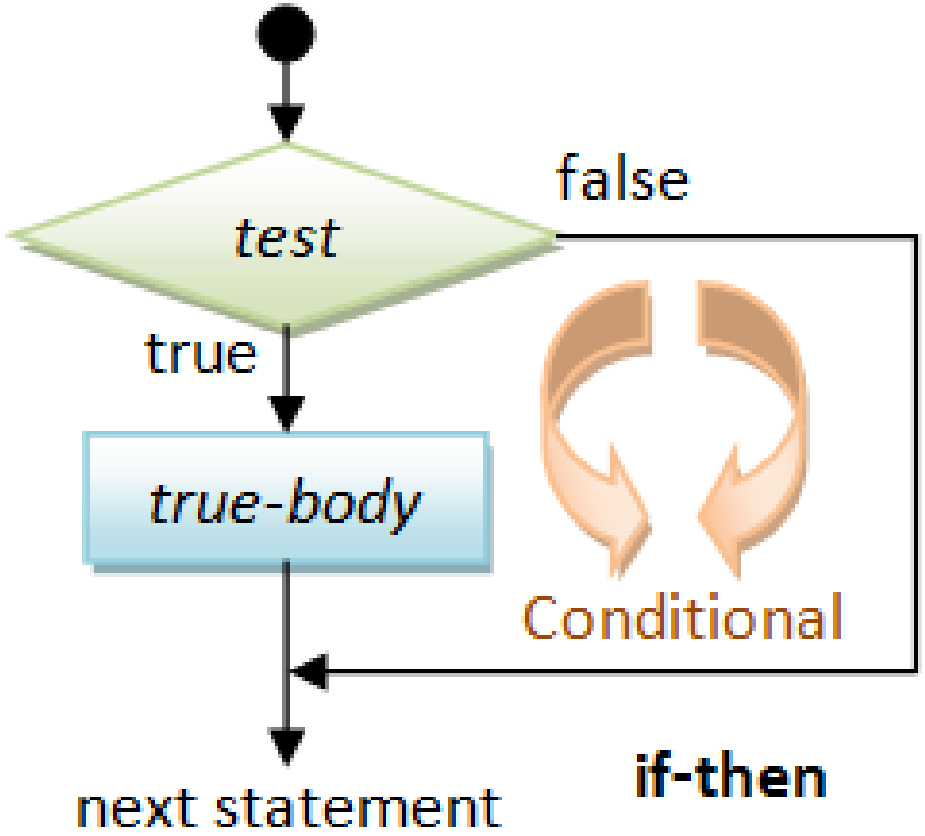
Sequence

- Statement are executed in the order in which they are specified in program.



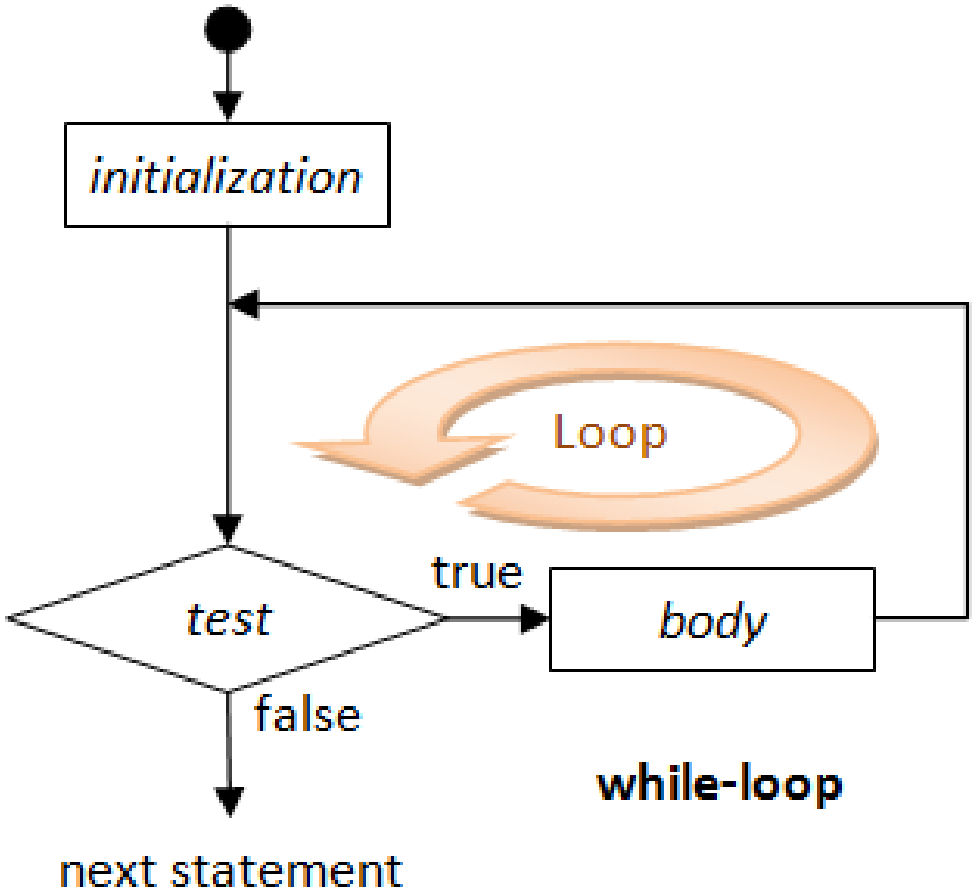
Selection

- Statement or set of statement executed on the basis of a condition(s).



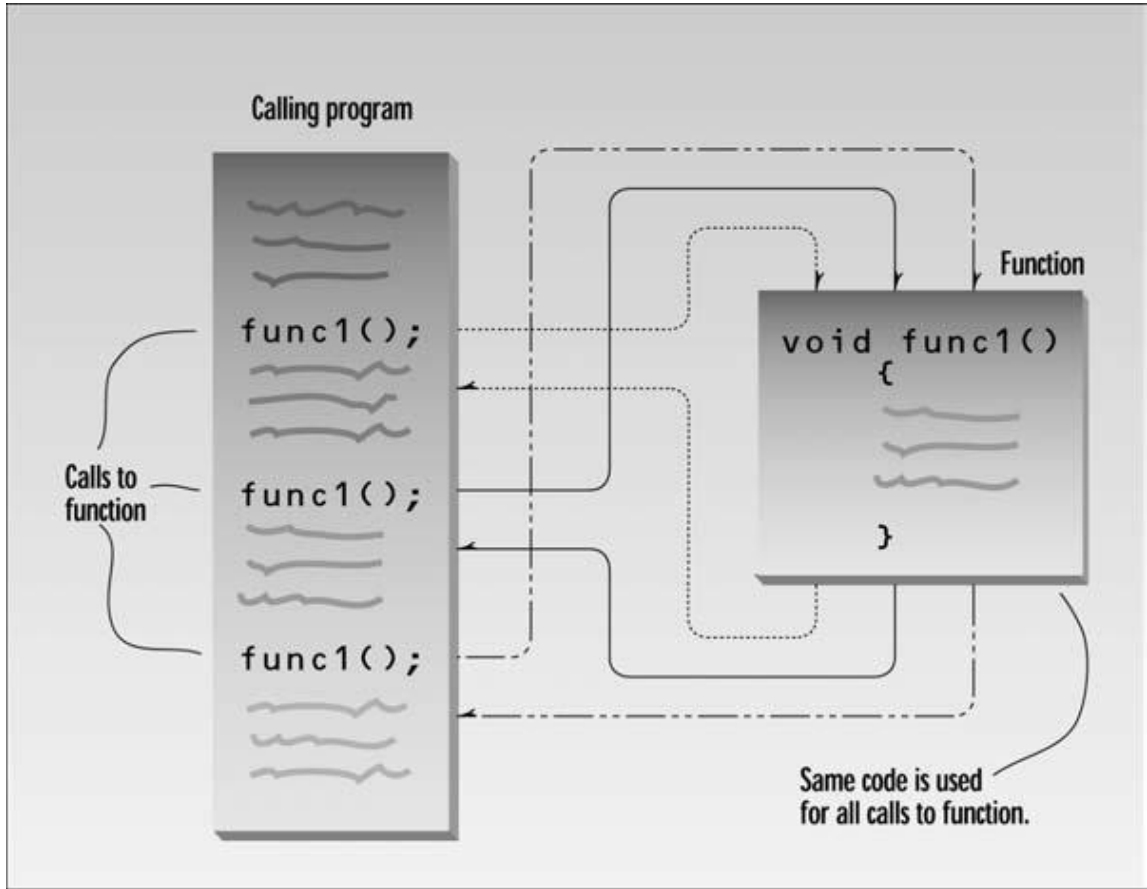
Repetition

- A repetition structure executes a statement or set statements repeatedly.



Function Call

- A type of statement that move the control to another block of code.



Selection

If Statement

- A conditional statement is a statement that specifies whether some associated statement(s) should be executed or not.

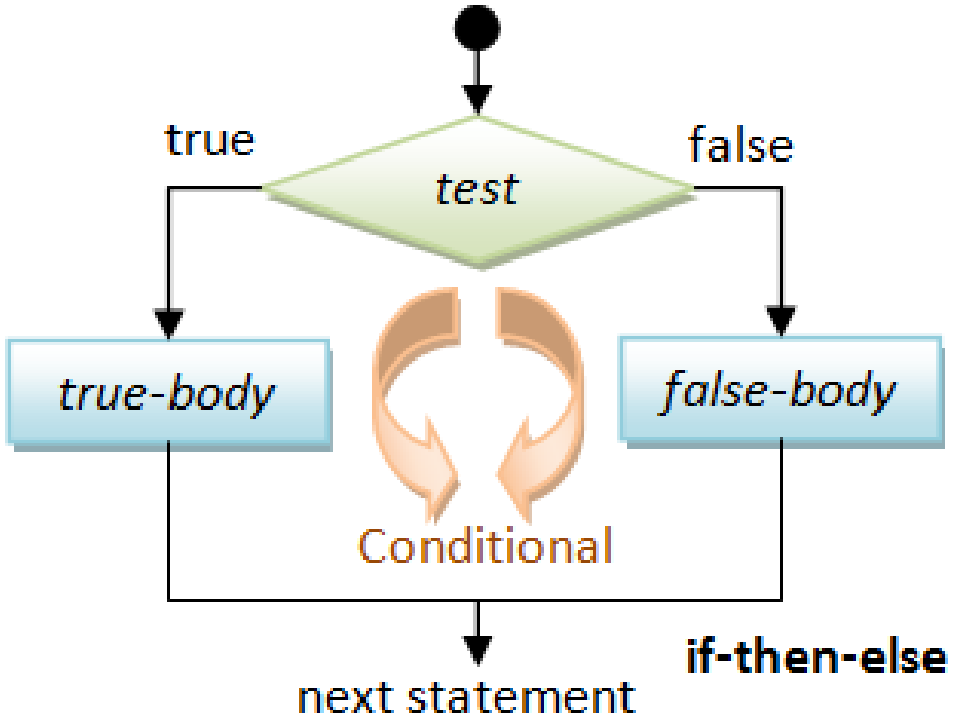
```
if (condition)
        true_statement;
```

- If statement is limited to only true case, what happed if the condition is false?

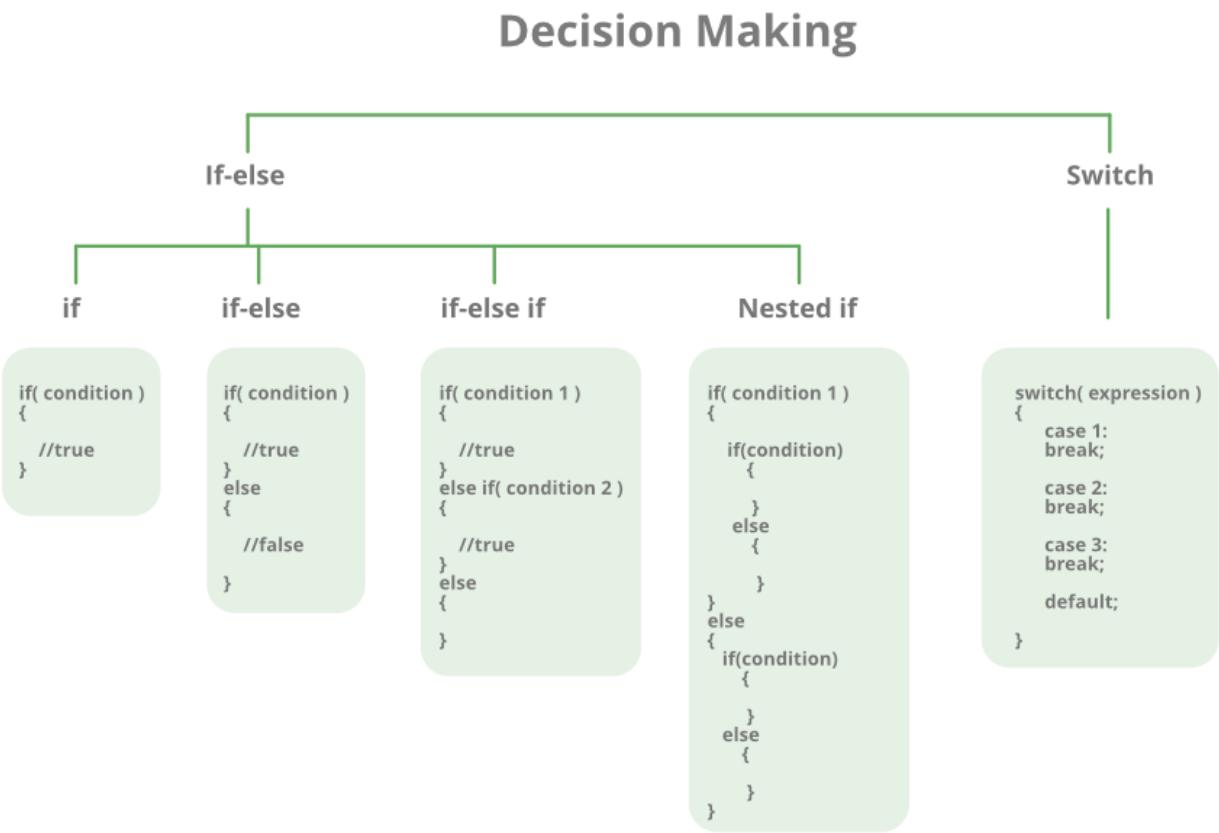
If-else

- Execute one block of code when condition is true and the other when condition is false.

```
if (condition)
        true_statement;
else
        false_statement;
```

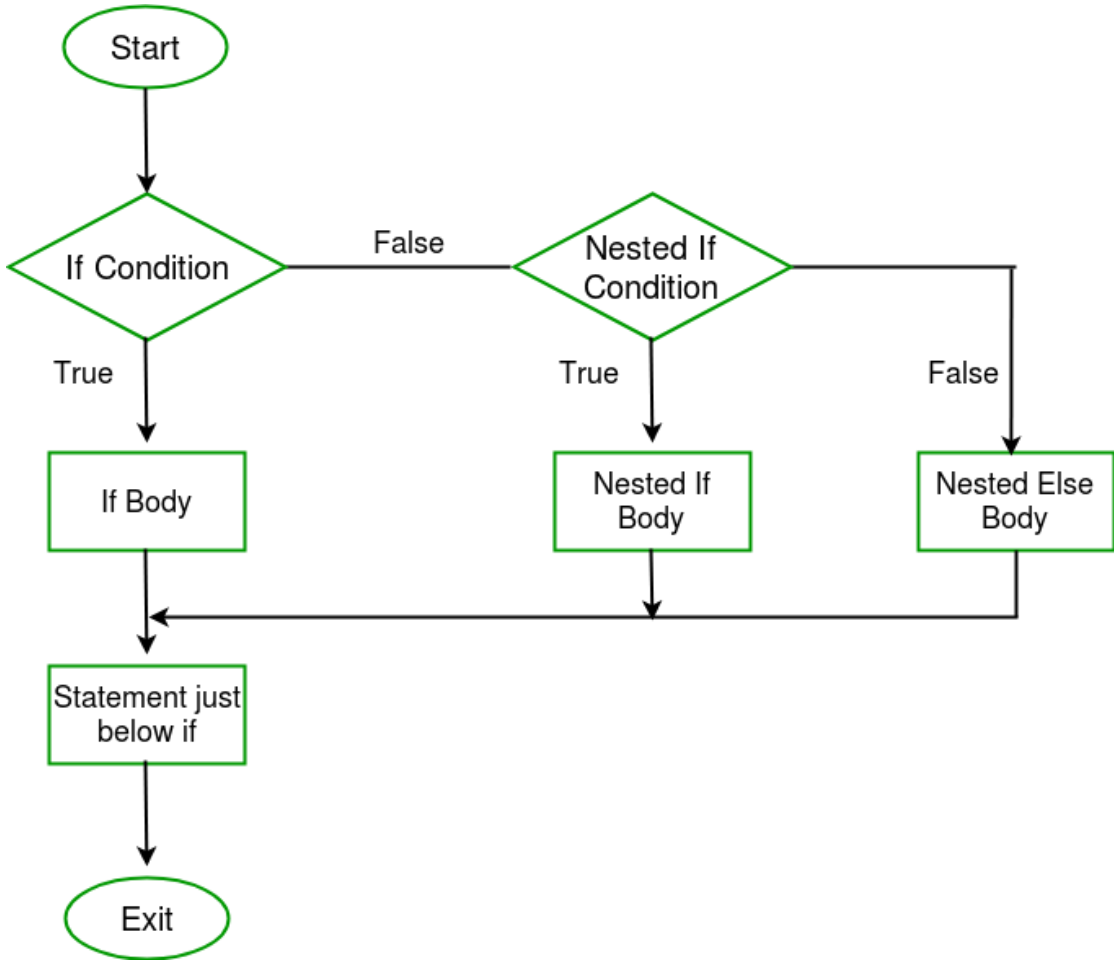


Multiple if-else

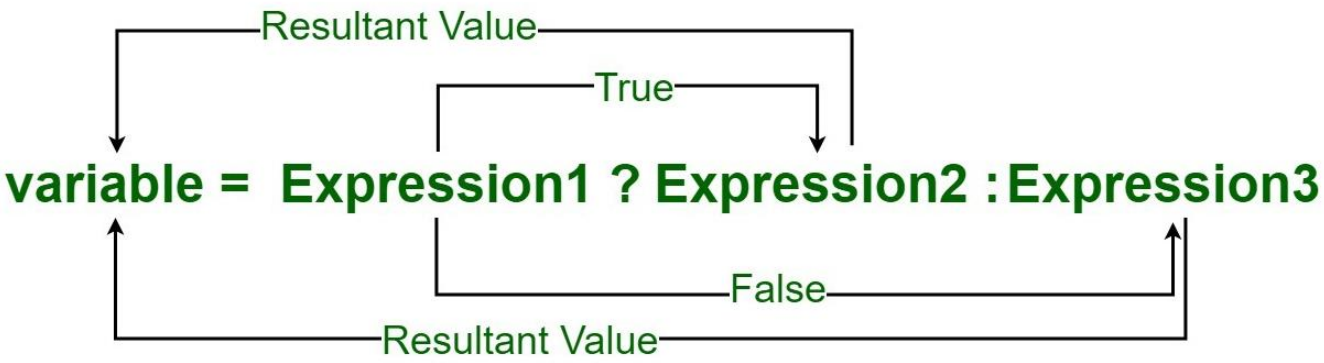


Nested If

- An if statement with-in if statement.



Conditional Operator



Switch

Switch case statements are a substitute for long if statements that compare a variable to several integral values:

- The switch statement is a multiway branch statement. It provides an easy way to dispatch execution to different parts of code based on the value of the expression.
- Switch is a control statement that allows a value to change control of execution.

Switch

```
switch (n)
{
    case 1: // code to be executed if n = 1;
        break;
    case 2: // code to be executed if n = 2;
        break;
    default: // code to be executed if n doesn't match any cases
}
```